Sürekli Zamanlı Fourier Transformu (Dönüşümü)

(Continuous-Time Fourier Transform)

Periyodik olmayan (aperiyodik) işaretler için Fourier dönüşümü tanımlıdır.

$$X(f) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

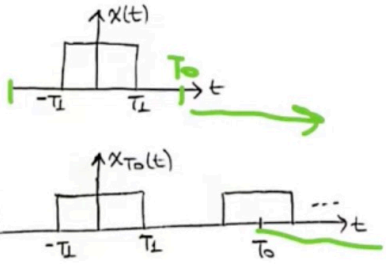
$$x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi ft}\, df$$

$$x(t) \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftarrows}} X(f)$$

⇒ ω notasyonu: $X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega \quad , \omega = 2\pi f = 2\pi/T$$

⇒ Periyodik olmayan bir işaretin belli bir T_0 aralığında Fourier serisi açılımında T_0 sonsuza götürüldüğünde ($T_0 \to \infty$) Fourier transformu elde edilir.



$T_0 \to \infty$ iken $x_{T_0}(t) = x(t)$ olur (herhangi bir sonlu t değeri için)

$$\lim_{T_0 \to \infty} x_{T_0}(t) = x(t)$$

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} x_k\, e^{jk\omega_0 t} \quad , \omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

$$x_k = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x_{T_0}(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T_0} \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t}\, dt$$

($x_{T_0}(t) = x(t)$ for $|t| < T_0/2$; 0 for $|t| > T_0/2$; $T_0 \to \infty$)

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}$$

$$x_k = \frac{1}{T_0} X(kf_0)$$

$$x_k = \frac{1}{T_0} X(k\omega_0)$$

df=1/T0=f0

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_0} X(kf_0)\, e^{j2\pi kf_0 t}$$

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_0} X(k\omega_0)\, e^{jk\omega_0 t}$$

$$\Delta f = \frac{1}{T_0}$$

$$\Delta\omega = \frac{2\pi}{T_0}$$

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} X(k\Delta f)\, e^{j2\pi k\Delta f t}\, \Delta f$$

$$x_{T_0}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\Delta\omega)\, e^{jk\Delta\omega t}\, \Delta\omega$$

$$x(t) = \lim_{T_0 \to \infty} x_{T_0}(t)$$

$$= \lim_{\Delta f \to 0} \sum_{k=-\infty}^{\infty} X(k\Delta f)\, e^{j2\pi k\Delta f t}\, \Delta f$$

$X(f)e^{j2\pi ft}$'nin altında kalan alan

$$x(t) = \lim_{\Delta\omega \to 0} \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\Delta\omega)\, e^{jk\Delta\omega t}\, \Delta\omega$$

$X(\omega)e^{j\omega t}$

$$x(t) = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi ft}\, df$$

$$x(t) = \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

⇒ Fourier dönüşümünün mevcut olma koşulları (Dirichlet condition)

1) Aperyodik $x(t)$ işareti mutlak integrallenebilir olmalıdır, $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

2) Herhangi bir sonlu t aralığında,

- sonlu sayıda maksimum ve minimum değere sahip olmalıdır.
- sonlu sayıda süreksizliğe ve bu noktalarda sonlu değere sahip olmalıdır.

⇒ Frekans spektrumları:

$X(f) = |X(f)|\, e^{j\theta_x(f)}$ → $|X(f)|$: genlik spektrumu

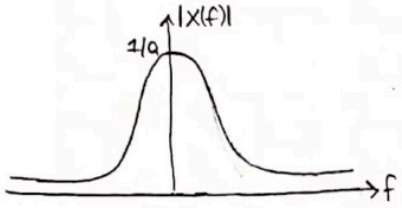
↳ $\theta_x(f)$: faz spektrumu

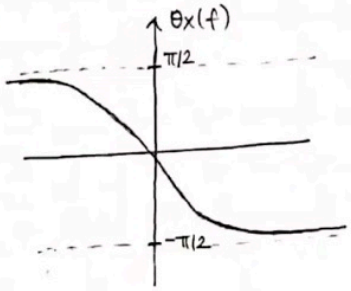
Ex: $x(t) = e^{-at}u(t)$, $a>0$ ise $X(f) = ?$ Frekans spektrumlarını çiziniz.

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt = \int_{0}^{\infty} e^{-at}\cdot e^{-j2\pi ft}dt = \int_{0}^{\infty} e^{-(a+j2\pi f)t}dt = \frac{e^{-(a+j2\pi f)t}}{-(a+j2\pi f)}\Big|_{0}^{\infty} = \frac{1}{a+j2\pi f} = \frac{1}{a+j\omega}$$

$$|X(f)| = \frac{1}{\sqrt{a^2+(2\pi f)^2}}$$

$$\theta_X(f) = -\tan^{-1}(2\pi f/a)$$





Ex: $x(t) = e^{-a|t|}$, $a>0$. ise $X(f) = ?$

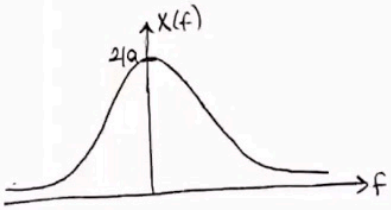
$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt = \int_{-\infty}^{0} e^{at}e^{-j2\pi ft}dt + \int_{0}^{\infty} e^{-at}e^{-j2\pi ft}dt$$

$$x(t) = \begin{cases} e^{at}, & t<0 \\ e^{-at}, & t>0 \end{cases}$$

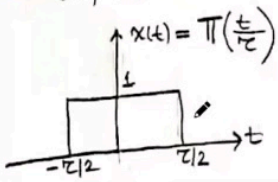
$$= \frac{e^{(a-j2\pi f)t}}{(a-j2\pi f)}\Big|_{-\infty}^{0} + \frac{e^{-(a+j2\pi f)t}}{}\Big|_{0}^{\infty} = \frac{1}{a-j2\pi f} + \frac{1}{a+j2\pi f} = \frac{2a}{a^2+(2\pi f)^2}$$

$$|X(f)| = \frac{2a}{a^2+(2\pi f)^2}$$

$$\theta_X(f) = 0$$



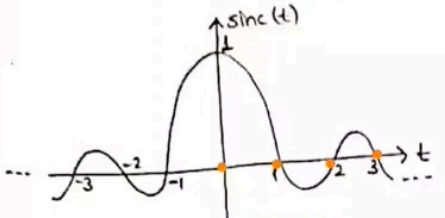
Ex: $x(t) = \begin{cases} 1, & |t| < \tau/2 \\ 0, & \text{dışında} \end{cases}$ dikdörtgen darbe işaretinin FT bulunuz.



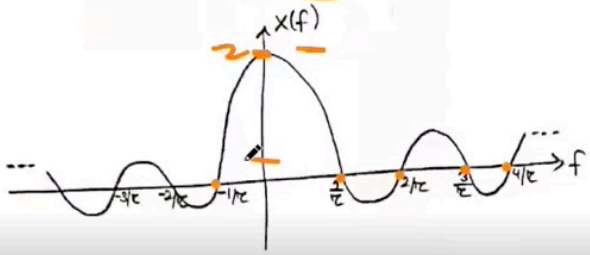
$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt = \int_{-\tau/2}^{\tau/2} e^{-j2\pi ft}\, dt = \frac{e^{-j2\pi ft}}{-j2\pi f}\Bigg|_{-\tau/2}^{\tau/2}$$

$$= \frac{-1}{j2\pi f}\left[e^{-j2\pi f\tau/2} - e^{+j2\pi f\tau/2} \right] = \frac{1}{\pi f}\sin(\pi f\tau)$$

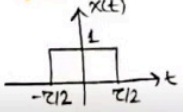
Reminder: $\operatorname{sinc}(t) = \begin{cases} \dfrac{\sin(\pi t)}{\pi t}, & t \neq 0 \\ 1, & t = 0 \end{cases}$



$$X(f) = \frac{1}{\pi f}\sin(\pi f\tau) = \tau\,\frac{\sin(\pi f\tau)}{\pi f\tau} = \tau\operatorname{sinc}(f\tau)$$



$$x(t) = \Pi\left(\frac{t}{\tau}\right) \overset{\mathcal{F}}{\longleftrightarrow} X(f) = \tau\operatorname{sinc}(f\tau)$$



Ex: $x(t) = \delta(t)$ ise $X(f) = ?$

$$X(f) = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi f \cdot 0} dt = \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$x(t) = \delta(t) \xleftrightarrow{\mathcal{F}} X(f) = 1$$



Ex: $X(f) = \delta(f - f_0)$ ise $x(t) = ?$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} \delta(f - f_0) e^{j2\pi ft} df = \int_{-\infty}^{\infty} \delta(f - f_0) e^{j2\pi f_0 t} df = e^{j2\pi f_0 t} \underbrace{\int_{-\infty}^{\infty} \delta(f - f_0) df}_{=1} = e^{j2\pi f_0 t}$$

Ex: $X(f) = \delta(f)$ ise $x(t) = ?$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} \delta(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} \delta(f) e^{j2\pi \cdot 0 \cdot t} df = \int_{-\infty}^{\infty} \delta(f) df = 1$$

$\delta(t) \xleftrightarrow{\mathcal{F}} 1$

$1 \xleftrightarrow{\mathcal{F}} \delta(f)$

$e^{j2\pi f_0 t} \xleftrightarrow{\mathcal{F}} \delta(f - f_0)$

$e^{-j2\pi f_0 t} \xleftrightarrow{\mathcal{F}} \delta(f + f_0)$

Periyodik işaretlerin Fourier Dönüşümü

$x(t) = x(t+T)$, $\forall t$ $\quad T = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$

$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} x_k e^{j2\pi k f_0 t} = \dots + x_{-1} e^{-j2\pi f_0 t} + x_0 + x_1 e^{j2\pi f_0 t} + x_2 e^{j2\pi 2 f_0 t} + \dots$$

$\omega_0 = 2\pi f_0$

$x_1 e^{j2\pi f_0 t} \xleftrightarrow{FT} x_1 \delta(f - f_0)$

$x_0 \xleftrightarrow{FT} x_0 \delta(f)$

$x_2 e^{j2\pi 2 f_0 t} \xleftrightarrow{FT} x_2 \delta(f - 2f_0)$

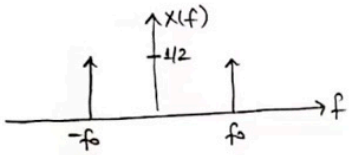
$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{j2\pi k f_0 t} \xleftrightarrow{FT} X(f) = \sum_{k=-\infty}^{\infty} x_k \delta(f - k f_0)$$

$$x_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

Ex: $x(t) = \cos 2\pi f_0 t$ ise $X(f) = ?$

$x(t) = \frac{1}{2}\left[e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}\right] \rightarrow x_1 = x_{-1} = 1/2$

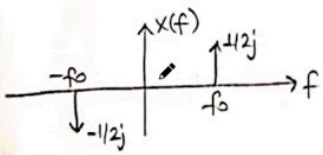
$X(f) = \frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$

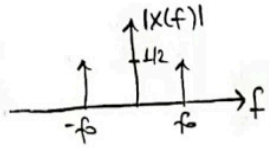


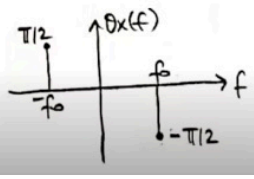
Ex: $x(t) = \sin 2\pi f_0 t$

$x(t) = \frac{1}{2j}\left[e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}\right]$ $\rightarrow x_1 = \frac{1}{2j} = \frac{-1}{2}j$ $\quad x_{-1} = \frac{-1}{2j} = \frac{1}{2}j$ $\quad \left.\begin{array}{l} |x_1| = |x_{-1}| = 1/2 \\ \theta_{x1} = -\pi/2 \quad \theta_{x-1} = \pi/2 \end{array}\right\}$

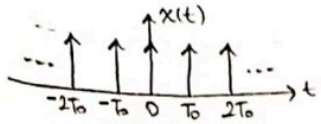
$X(f) = \frac{1}{2j}\left[\delta(f - f_0) - \delta(f + f_0)\right]$







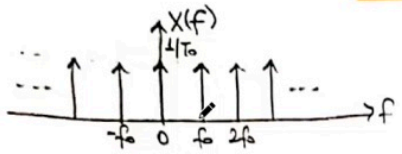
Ex: $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0)$ dürtü katarı işaretinin FT bulunuz.



Reminder: $x_k = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jk\omega_0 t} = \frac{1}{T_0}$



$$X(f) = \sum_{k=-\infty}^{\infty} x_k\, \delta(f - kf_0) = \frac{1}{T_0}\sum_{k=-\infty}^{\infty} \delta(f - kf_0)$$



Sürekli Zamanlı Fourier Transformunun Özellikleri

$x(t) \xleftrightarrow{\mathcal{F}} X(f)$, $x(t)$: aperiyodik işaret

$X(f) = \mathcal{F}\{x(t)\}$

1) Lineerlik: $\alpha x_1(t) + \beta x_2(t) \xleftrightarrow{\mathcal{F}} \alpha X_1(f) + \beta X_2(f)$

2) Zamanda öteleme: $x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j2\pi f t_0} X(f)$

$$\mathcal{F}\{x(t-t_0)\} = \int_{-\infty}^{\infty} x(t-t_0)\, e^{-j2\pi f t}\, dt = \int_{-\infty}^{\infty} x(\tau)\, e^{-j2\pi f(\tau + t_0)}\, d\tau = e^{-j2\pi f t_0} \underbrace{\int_{-\infty}^{\infty} x(\tau)\, e^{-j2\pi f \tau}\, d\tau}_{X(f)}$$

($t - t_0 = \tau$)

Aperiyodik bir $x(t)$ işareti zamanda ötelendiğinde genlik spektrumu değişmez, faz spektrumu $-2\pi f t_0$ kadar ötelenir.

3) Frekansta öteleme: $x(t)\, e^{j2\pi f_0 t} \xleftrightarrow{\mathcal{F}} X(f - f_0)$

İspat: $\mathcal{F}\{x(t)\, e^{j2\pi f_0 t}\} = \int_{-\infty}^{\infty} x(t)\, e^{j2\pi f_0 t}\, e^{-j2\pi f t}\, dt = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi (f - f_0) t}\, dt = X(f - f_0)$

$x(t)$ işareti $e^{j2\pi f_0 t}$ ile çarpıldığında frekans spektrumu f_0 kadar ötelenir.

4) Modülasyon: $x(t)\cos 2\pi f_0 t \xleftrightarrow{\mathcal{F}} \frac{1}{2}\left[X(f - f_0) + X(f + f_0)\right]$

$$x(t)\sin 2\pi f_0 t \xleftrightarrow{\mathcal{F}} \frac{1}{2j}\left[X(f - f_0) - X(f + f_0)\right] = \frac{1}{2}\left[X(f - f_0)\, e^{-j\pi/2} + X(f + f_0)\, e^{j\pi/2}\right]$$

5) Zamanda ölçeklendirme: $x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} X(f/a)$, $a \neq 0$ → $a>1$ işaret frekansta (spektrum) genişler.

$0<a<1$ işaret frekansta daralır.

6) Duality: $x(t) \xleftrightarrow{\mathcal{F}} X(f)$

$X(t) \xleftrightarrow{\mathcal{F}} x(-f)$

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İspat: $x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$

$x(-t) = \int_{-\infty}^{\infty} X(f) e^{-j2\pi ft} df \Rightarrow x(-f) = \int_{-\infty}^{\infty} X(t) e^{-j2\pi ft} dt$

$t \to f$

X(t)

7) Türev alma: $\frac{dx(t)}{dt} \xleftrightarrow{\mathcal{F}} j2\pi f X(f)$

Ex: $x(t) = A\Pi(t/\tau) \xleftrightarrow{\mathcal{F}} X(f) = A\tau \operatorname{sinc}(f\tau)$ ise $x(t) = \operatorname{sinc}(t\tau)$ işaretinin FT=?

Reminder: $x(t) \longleftrightarrow X(f)$, $X(f) \longleftrightarrow x(-t)$ } Duality

$x(t) = A\tau \operatorname{sinc}(t\tau) \xleftrightarrow{\mathcal{F}} X(f) = A\Pi(-f/\tau) = A\Pi(f/\tau)$

↑ Çift fonksiyon

$x(t) = \operatorname{sinc}(t\tau) \xleftrightarrow{\mathcal{F}} X(f) = \frac{1}{\tau}\Pi(f/\tau)$



Ex: $x(t) = \frac{2}{1+t^2}$ ise $X(f) = ?$

Reminder: $e^{-a|t|} \longleftrightarrow \frac{2a}{a^2+(2\pi f)^2}$

$x(t) = e^{-|t|} \longleftrightarrow X(f) = \frac{2}{1+(2\pi f)^2}$

$x(t) = \frac{2}{1+t^2} \longleftrightarrow X(f) = e^{-|-f|} = e^{-|f|}$

Ex: $x(t) = e^{-|t-t_0|}$ ise $X(f) = ?$

Reminder: $x(t-t_0) \longleftrightarrow e^{-j2\pi f t_0} X(f)$ ve $e^{-|t|} \longleftrightarrow \frac{2a}{a^2+(2\pi f)^2}$

$x(t) = e^{-|t-t_0|} \xleftrightarrow{\mathcal{F}} X(f) = \frac{2a}{a^2+(2\pi f)^2} e^{-j2\pi f t_0}$

$\longrightarrow |X(f)| = \frac{2a}{a^2+(2\pi f)^2}$

$\longrightarrow \theta_x(f) = -2\pi f t_0$

